

Linear Algebra Lesson 2: Fields

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For our second correspondence, I will begin delving into the basics of abstract linear algebra. When I first learned these concepts, I thought it was super interesting to see how a lot of elementary school math (like basic arithmetic) was based in these more abstract concepts that you don't learn until higher-level math. But a lot of these concepts are the underlying basis of super simple math! For this letter, we will be discussing the concept of fields. As a side note, from my experience studying math, a lot of it is actually learning definitions. These definitions help conceptualize new components that can then be used in the logic of proofs.

1. Definition: Field (of Scalars)

A set of numbers that comes with operations of addition (denoted with $+$) and multiplication (denoted with $\cdot$ or $*$) is called a **field** (of scalars), denoted $\mathbb{F}$, if it satisfies the following properties:

(a) Commutative Property:

$$x + y = y + x \text{ and} \\ x \cdot y = y \cdot x \text{ for all } x, y \in \mathbb{F}$$

(b) Associative Property:

$$(x + y) + z = x + (y + z) \text{ and} \\ (x \cdot y) \cdot z = x \cdot (y \cdot z) \text{ for all } x, y, z \in \mathbb{F}$$

(c) Distributive Property:

$$x \cdot (y + z) = x \cdot y + x \cdot z \text{ for all } x, y, z \in \mathbb{F}$$

(d) Identities:

There exists the distinct elements $0, 1 \in \mathbb{F}$, also known as identities, such that

i. Additive Identity (0): $x + 0 = x$ for all $x \in \mathbb{F}$

ii. Multiplicative Identity (1): $x \cdot 1 = x$ for all $x \in \mathbb{F}$

(e) Inverses:

i. Additive Inverse: For all $x \in \mathbb{F}$, there exists a unique $y \in \mathbb{F}$ such that $x + y = 0$

ii. Multiplicative Inverse: For all nonzero $x \in \mathbb{F}$, there exists a unique $y \in \mathbb{F}$ such that $x \cdot y = 1$

2. Examples and Non-examples of Fields:

Ultimately, we have this definition of fields because there are certain theorems or propositions that only apply to fields. So, it is useful to know if a set of numbers can be considered a field or not. The best way to check if something is a field is to check that all 5 properties listed above in the definition (Commutative, Associative, Distributive, Identities, and Inverses) apply. Also a side note — I don't know how familiar you are with inverses, but they confused me a little bit when I first learned them. But basically the additive inverse is kind of like subtraction, and the multiplicative inverse is kind of like division. So, for example, the additive inverse of x is $-x$, and the multiplicative inverse of x is $1/x$. That way, instead of considering addition, subtraction, multiplication, and division, we just need to consider addition (where subtraction is just adding negative numbers, or additive inverses) and multiplication (where division is just multiplying by fractions, or multiplicative inverses).

(a) Examples of Fields:

- i. $\mathbb{R}$: Real numbers
- ii. $\mathbb{C}$: Complex numbers
- iii. $\mathbb{Q}$: Rational numbers

(b) Non-Examples of fields:

- i. $\mathbb{N}$: Natural numbers $\{1, 2, 3, \dots\}$

A. This is because the natural numbers do not satisfy the additive inverse property, which states that for all $x \in \mathbb{F}$, there exists a unique $y \in \mathbb{F}$ such that $x + y = 0$. However, consider the element $x = 2$ from $\mathbb{N}$. There does not exist a y from $\mathbb{N}$ that is the additive inverse of x , or 2 .

B. For example, -2 is the additive inverse of 2 , but -2 is not from the natural numbers. Thus, 2 does not have an additive inverse inside the set of natural numbers. This means the set of natural numbers is not a field.

- ii. $\mathbb{Z}$: Integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$

A. This is because the integers do not satisfy the multiplicative inverse property, which states that for all nonzero $x \in \mathbb{F}$, there exists a unique $y \in \mathbb{F}$ such that $x \cdot y = 1$. Consider $x = 2$, which is a nonzero integer. There does not exist a y from $\mathbb{N}$ that is the multiplicative inverse of x , or 2 .

B. For example, $\frac{1}{2}$ is the multiplicative inverse of 2 , but $\frac{1}{2}$ is not an integer. Thus, 2 does not have a multiplicative inverse inside the set of integers. This means the set of integers is not a field.

3. Subtraction, Division, Exponents

Once we have properties (a) - (e) from our 1. Definition: Field, we can

carry out subtraction $(-)$ and division $(/)$ in $\mathbb{F}$ as well, as I briefly touched on above.

(a) Subtraction:

- i. For every $x \in \mathbb{F}$, its additive inverse is denoted $-x \in \mathbb{F}$
- ii. For every $x, y \in \mathbb{F}$, we use $x - y$ to denote $x + (-y)$

(b) Division:

- i. For every nonzero $x \in \mathbb{F}$, its multiplicative inverse is denoted $\frac{1}{x} \in \mathbb{F}$
- ii. For every $x, y \in \mathbb{F}, y \neq 0$, we use $\frac{x}{y}$ to denote $x \cdot \frac{1}{y}$

(c) Exponents:

- i. For every $x \in \mathbb{F}, m \in \mathbb{N}$, we use x^m to denote x multiplied by itself m times, $x \cdot \dots \cdot x$

4. More Examples of Fields

Since the examples of fields we gave are pretty common sets of numbers, you might wonder why we even defined this term "field" when we can just call the set of "real numbers" or "rational numbers" or something. Well, that's because there are more examples of fields! Below, we show a fun example of a field called $\mathbb{F}_2 = \{0, 1\}$.

I don't know if you have heard of binary arithmetic, but this field is basically like the binary system. Our regular numbers are in base 10, but binary arithmetic is in base 2. Computers use binary language.

I like this one joke about it: There are 10 types of people. Those that understand binary numbers, and those that don't. (This is because 10 equals 2 in binary language.) Anyways, if you have interest in the concept of binary numbers I can add a little section in our next letter, as it can be pretty relevant to abstract algebra. Anyways, let us prove below that $\mathbb{F}_2$ is a field.

(a) $\mathbb{F}_2 = \{0, 1\}$. Below in our tables, we define addition and multiplication within this field. (E.g., $1 + 1 = 0$ in $\mathbb{F}_2$)

Table 1: Addition in $\mathbb{F}_2$

+	0	1
0	0	1
1	1	0

Table 2: Multiplication in $\mathbb{F}_2$

$\cdot$	0	1
0	0	0
1	0	1

(b) Now, we would prove it is a field by going through the 5 properties and seeing $\mathbb{F}_2$ satisfies all of them. It is a bit tedious to prove it by hand completely, given the current tools in our toolkit, but I will at least provide an outline for each property.

i. Commutativity

A. Additive commutativity: $0 + 1 = 1 = 1 + 0$ QED.

B. Multiplicative commutativity: $0 \cdot 1 = 0 = 1 \cdot 0$ QED.

ii. Associativity

A. Additive associativity

$0 + (1 + 0) = 1 = (0 + 1) + 0$ is true,

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$1 + (0 + 1) = 0 = (1 + 0) + 1$ is true,

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$0 + (1 + 1) = 0 = (0 + 1) + 1$ is true,

$1 + (1 + 1) = 1 = (1 + 1) + 1$ is true,

$0 + (0 + 0) = 0 = (0 + 0) + 0$ is true.

Since these are all the possible associative combinations, and they are all true based on our definition of addition in $\mathbb{F}_2$, we have proven additive associativity applies in $\mathbb{F}_2$.

I will also say that in the future, you do not always do proofs by showing each possible combination by hand. You usually show it by using variables and calling back on proven theorems. But right now, because we have not proven many theorems yet, and because $\mathbb{F}_2$ is such a small field, it is not too hard to show every single permutation.

B. Multiplicative associativity

$0 \cdot (1 \cdot 0) = 0 = (0 \cdot 1) \cdot 0$ is true,

$1 \cdot (0 \cdot 0) = 0 = (1 \cdot 0) \cdot 0$ is true,

$0 \cdot (0 \cdot 1) = 0 = (0 \cdot 0) \cdot 1$ is true,

$1 \cdot (0 \cdot 1) = 0 = (1 \cdot 0) \cdot 1$ is true,

$1 \cdot (1 \cdot 0) = 0 = (1 \cdot 1) \cdot 0$ is true,

$0 \cdot (1 \cdot 1) = 0 = (1 \cdot 1) \cdot 1$ is true,

$1 \cdot (1 \cdot 1) = 1 = (1 \cdot 1) \cdot 1$ is true,

$0 \cdot (0 \cdot 0) = 0 = (0 \cdot 0) \cdot 0$ is true.

Since these are all the possible associative combinations, and they are all true based on our definition of multiplication in

$\mathbb{F}_2$, we have proven multiplicative associativity applies in $\mathbb{F}_2$.

iii. Distributivity

A. I will not list out every single possible permutation as I did for associativity, but I will show an example:

$$1 \cdot (1 + 0) = 1 = 1 \cdot 1 + 1 \cdot 0 \text{ is true.}$$

Then, to finish proving distributivity, you would check to see every permutation is also true.

iv. Additive Inverses

We need to show each element 0,1 has an additive inverse in $\mathbb{F}_2$.

A. $1 + 1 = 0$, hence, 1 is the additive inverse of 1.

B. $0 + 0 = 0$, hence, 0 is the additive inverse of 0.

Nice! We have shown both elements have an additive inverse.

v. Multiplicative Inverses

We need to show each nonzero element of $\mathbb{F}_2$ has a multiplicative inverse in $\mathbb{F}_2$.

A. $1 \cdot 1 = 1$, hence, 1 is the additive inverse of 1.

Since 1 is the only nonzero element of $\mathbb{F}_2$, we have shown all nonzero elements have a multiplicative inverse!

(c) **Exercises**

Below are some exercises regarding fields.

- i. We have shown all of the above using $\mathbb{F}_2$. Now, consider $\mathbb{F}_3$ which has the elements $\{0,1,2\}$. Fill in the following tables of additive and multiplicative inverses. (Hint: 0 does not have a multiplicative inverse, so that one box will be blank.)

Table 3: Additive Inverses in $\mathbb{F}_2$

x	0	1	2
-x			

Table 4: Multiplicative Inverses in $\mathbb{F}_2$

x	0	1	2
1/x			

- ii. Using $\mathbb{F}_3$, given that you have just filled out the additive and multiplicative inverse tables, compute

$$\frac{2^3 + 1}{2} \in \mathbb{F}_3$$

- iii. Complex numbers are denoted by $a + bi$, where $a, b \in \mathbb{R}$ (a, b are real numbers), and $i = \sqrt{-1}$. Complex numbers are a field. With that in mind, compute (in terms of a, b, i)

$$z^3 \text{ where } z = a + bi$$

iv. Show

$$w^3 = 1 \text{ where } w = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

v. Find additive and multiplicative inverses for

$$w = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$